STATISTICS

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1. What is Statistics?

Statistics is the science of collecting, organizing, analyzing, and interpreting data to make better decisions or understand patterns. It helps us make sense of numbers and draw conclusions from data.

2. What is the central limit theorem?

The Central Limit Theorem says that if we take a large number of random samples from any population, the sampling distribution of the sample mean will be approximately normal, no matter the shape of the original population.

3. What is a 5-Number Summary ?

A 5-number summary gives a quick overview of a dataset. It includes:

1. **Minimum** – the smallest value
2. **Q1 (First Quartile)** – 25% of the data is below this
3. **Median (Q2)** – the middle value
4. **Q3 (Third Quartile)** – 75% of the data is below this
5. **Maximum** – the largest value

It’s often used to create boxplots.

4. What is hypothesis testing?

It’s a method to check if a claim about a population is likely true or not, using sample data. We set up two statements:

- Null hypothesis (H_0): no effect or no difference
- Alternate hypothesis (H_1): what we want to prove
  Then we use data and a test statistic to decide whether to reject H_0 or not

5. What is the law of large numbers?

The Law of Large Numbers says that as we collect more and more data (or increase the sample size), the sample mean will get closer to the true population mean. In short, more data gives more accurate results.

6. Explain p-value with help of example?

The p-value tells us how likely it is to get our sample result just by chance, assuming the null hypothesis is true.

A small p-value (like less than 0.05) means the result is unlikely by chance, so we reject the null hypothesis.

Let's say a company claims their battery lasts **10 hours**. You test a sample and find the average is **8.5 hours**. Now you want to check if this difference is just due to random chance or if the battery really doesn't last 10 hours.

You set up:

- **H_0**: Battery lasts 10 hours
  H_1: Battery does not last 10 hours

After running the test, suppose you get a **p-value of 0.01**.

This means:
There's a **1% chance** of getting a result like 8.5 hours **if** the battery really does last 10 hours.

Since 0.01 is small (less than 0.05), you **reject H_0** and say the battery likely doesn't last 10 hours.

7. Explain type 1 and type 2 errors?

In hypothesis testing:

- **Type 1 Error** happens when we reject the null hypothesis even though it is true.
  👉 Like saying someone is guilty when they're actually innocent.
  Type 2 Error happens when we fail to reject the null hypothesis even though it's false.
  👉 Like saying someone is innocent when they're actually guilty.

In short:

Type 1 = False positive
Type 2 = False negative

8. What is the difference between a sample and a population?

- **Population** is the whole group we want to study or make conclusions about.
  👉 Example: All students in India.

- **Sample** is a smaller part taken from the population for analysis.
  👉 Example: 200 students selected from different colleges.

We use samples because studying the entire population is often not possible.

9.What are the different kinds of variables or levels of measurement?

There are **four types of measurement levels**:

1. **Nominal** – Categories with no order.
   👉 Example: Gender, colors, city names
2. **Ordinal** – Categories with a clear order, but no fixed gap.
   👉 Example: Rankings (1st, 2nd, 3rd), satisfaction levels
3. **Interval** – Ordered, with equal gaps, but no true zero.
   👉 Example: Temperature in Celsius
4. **Ratio** – Like interval, but has a true zero.
   👉 Example: Age, height, weight, income

10. What is the difference between Descriptive and Inferential Statistics?

Descriptive Statistics is about **summarizing** and **presenting** data using numbers like mean, median, standard deviation, or graphs.
👉 Example: Average marks of a class.
Inferential Statistics is about using a **sample** to make **conclusions or predictions** about a population.
👉 Example: Predicting election results based on a survey.

11. What is sampling and Its Types?

Sampling is the process of selecting a small group (sample) from a larger group (population) to study and draw conclusions.

Types of Sampling:

1. **Random Sampling** – Every individual has an equal chance.
   👉 Like picking names from a hat.
2. **Systematic Sampling** – Select every k^{th} item.
   👉 Like every 10th person on a list.
3. **Stratified Sampling** – Divide population into groups (strata) and take samples from each.
   👉 Like sampling students from each class.
4. **Cluster Sampling** – Divide into clusters, randomly choose some clusters, then study everyone in those.
   👉 Like picking 3 colleges and surveying all students in them.
5. **Convenience Sampling** – Choose whoever is easiest to reach.
   👉 Example: Asking friends for a survey.
6. **Judgmental/Purposive Sampling** – Choose people based on your knowledge or judgment.
   👉 Example: Interviewing experts only.
7. **Snowball Sampling** – Existing subjects refer to new ones.
   👉 Example: Used in hidden or hard-to-reach groups.

8. **Quota Sampling** – Like stratified, but not random. You fill a quota from each group.
   👉 Example: 50 males and 50 females, but chosen by convenience.

12. In what situation would the median be a more suitable measure compared to the mean?

When the data are skewed, the median is more useful because the mean will be distorted by outliers.

13.What does standard deviation mean?

Standard deviation tells us how spread out the data is from the mean.

A **small standard deviation** means the values are close to the mean.
A **large standard deviation** means the values are more spread out.

It helps understand how consistent or varied the data is.

14. Explain correlation and causation with examples?

Correlation means two variables move together — when one changes, the other tends to change too.
👉 Example: Height and weight — taller people often weigh more.

Causation means one variable **actually causes** the change in the other.
👉 Example: Studying more **causes** better exam scores.

⚠️ Important: Correlation doesn't always mean causation.
👉 Ice cream sales and drowning both go up in summer — they're correlated, but one doesn't cause the other.

15. What is standard error and margin of error?

Standard Error (SE):
It measures how much the sample mean is expected to vary from the true population mean.
👉 Smaller SE means the sample mean is more reliable.

Margin of Error (MoE):
It tells us how much we expect the results to differ from the true value — it's based on the standard error and confidence level.
👉 Example: If survey says 60% ± 3%, the margin of error is 3%.

16. What is the meaning of Central Tendency?

Central tendency refers to the statistical measure that represents the center or typical value of a dataset. It helps describe where most data points cluster and includes measures like the mean, median, and mode.

17. How do you define Normal Distribution?

A normal distribution, also known as a Gaussian distribution, is a symmetric probability distribution characterized by a bell-shaped curve. In a normal distribution, data is centered around a mean, and its variability is described by a standard deviation.

18. How do you define empirical rule?

The empirical rule, also known as the 68-95-99.7 rule, is a statistical guideline that applies to approximately normal distributions. It states that:

- About 68% of the data falls within one standard deviation of the mean.

- Approximately 95% of the data falls within two standard deviations of the mean.

- Roughly 99.7% of the data falls within three standard deviations of the mean.

19. What Is the Confidence Interval?

A confidence interval is a statistical range or interval that estimates the range within which a population parameter, such as a mean or proportion, is likely to fall, with a specified level of confidence.

Example: If we say the average height is **165 cm ± 2 cm** with 95% confidence, it means we're 95% sure the true average is between **163 and 167 cm**.

20. When should you use a t-test vs a z-test?

You should use a t-test when:

-The sample size is small (typically $n < 30$).

- The population standard deviation is unknown and has to be estimated from the sample data.

You should use a z-test when:

- The sample size is large (typically $n \geq 30$).

- The population standard deviation is known or when you are comparing sample means to a known population mean.

21. What is an outlier? How can outliers be determined in a dataset?

An outlier is an observation or data point that significantly differs from the majority of the data in a dataset. Outliers can be determined in a dataset using statistical techniques such as the Interquartile Range (IQR) method, z-scores, or visualization methods like box plots and scatter plots. Outliers are data points that fall below the lower bound or above the upper bound defined by these methods

22. What is the Pareto principle?

The **Pareto Principle**, also called the **80/20 rule**, says that **80% of results come from 20% of causes**.

👉 Example: In sales, 80% of revenue might come from 20% of customers.

It's used to focus on the most important factors that have the biggest impact.

23. What is Bessel's correction?

Bessel's correction is used when we calculate the sample variance or standard deviation.

Instead of dividing by **n**, we divide by **n - 1** to make the estimate more accurate for the population.

👉 It corrects the bias in the estimation because a sample usually underestimates the true population variance.

24. What are measures of central tendency? How are they different?

Measures of central tendency are numbers that describe the center or average of a dataset. The three main types are:

1. **Mean** – The average (add all values and divide by count).
   👉 Affected by outliers.
2. **Median** – The middle value when data is sorted.
   👉 Not affected by outliers.
3. **Mode** – The value that appears most often.
   👉 Useful for categorical data.

   Difference:

- **Mean** is best for symmetric data.
- **Median** is better when data has outliers or is skewed.
- **Mode** is used when we want the most common value.

25. What are measures of dispersion? Why are they important?

Measures of dispersion show how spread out or varied the data is. Common types are:

1. **Range** – Difference between highest and lowest value
2. **Variance** – Average of squared differences from the mean
3. **Standard Deviation** – Square root of variance
4. **IQR (Interquartile Range)** – Spread of the middle 50% (Q3 − Q1)

Why is it important?
They help us understand how consistent or scattered the data is. Two datasets can have the same average but very different spreads.

26. Explain the concept of a probability distribution. Can you name some common distributions?

A **probability distribution** is a mathematical function that describes the likelihood of different outcomes in a random experiment. It specifies how probabilities are distributed over the possible values of a random variable.

1. Discrete Probability Distributions

- **Definition**: Distributions for random variables that take on a countable number of distinct values.

Common Distributions:

1. **Bernoulli Distribution:**
   - Models a single trial with two outcomes (success/failure).
   - Example: Tossing a coin once.
   - $$P(X = x) = p^x(1-p)^{1-x}, \quad \text{for } x \in \{0, 1\}$$
   - x is the outcome (0 or 1)
   - p is the probability of success (i.e., P(X=1)=pP(X = 1) = pP(X=1)=p)
   - 1−p is the probability of failure

2. **Binomial Distribution**:
   - Models the number of successes in a fixed number of independent trials.
   - Example: Number of heads in 10 coin tosses.
   - $$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- X = number of successes
- n = number of trials
- k = number of successes in those trials
- p = probability of success

3. **Poisson Distribution**:
    - Models the number of events occurring in a fixed interval of time or space.
    - Example: Number of emails received in an hour.
    - $$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$
    - X = number of events
    - k = actual number of occurrences (0, 1, 2, ...)
    - λ = average rate (mean number of occurrences in a fixed interval)

4. **Geometric Distribution**:
    - Models the number of trials until the first success.
    - Example: Number of attempts to hit a target.
    $$P(X = k) = (1 - p)^{k-1} \cdot p$$
    - X = number of trials until the **first success**
    - k = trial number where first success occurs (k = 1, 2, 3, ...)
    - p = probability of success in each trial
    - 1−p = probability of failure

2. Continuous Probability Distributions

- **Definition**: Distributions for random variables that take on an infinite number of values within a range.

Common Distributions:

1. **Normal Distribution**:
    - Bell-shaped curve, symmetric about the mean.
    - Example: Heights of people, IQ scores.
    - $$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

2. **Exponential Distribution**:

- Models the time between events in a Poisson process.
- Example: Time between arrivals of buses.
- $$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$
- x = time or distance until an event happens
- λ = rate parameter (average number of events per unit time)

3. **Uniform Distribution**:
   - All outcomes in a range are equally likely.
   - Example: Rolling a fair die (if treated as continuous).
   - $$f(x) = \frac{1}{b-a}, \quad \text{for } a \leq x \leq b$$
   - a = minimum value
   - b = maximum value
   - x = value in the interval [a,b]

27. What is the difference between descriptive and inferential statistics?

Descriptive Statistics is about **summarizing** and **describing** the data you have.
👉 Example: Mean, median, charts, standard deviation of a dataset.

Inferential Statistics is about **making predictions or decisions** about a population using sample data.
👉 Example: Estimating average income of a city using a survey of 500 people.

In short:

Descriptive = What the data shows

Inferential = What we can conclude beyond the data

28.Define probability. What are its key properties?

Probability is a measure of how likely an event is to happen, and it always lies between **0 and 1**.

- 0 means the event is impossible.
- 1 means the event is certain.

29. What is the difference between independent and dependent events?

Independent events are events where the outcome of one does **not** affect the other.
👉 Example: Tossing a coin and rolling a dice — one doesn't impact the other.

Dependent events are events where the outcome of one **does affect** the other.
👉 Example: Drawing two cards from a deck **without** replacement — the first draw affects the second.

In short:

- **Independent**: No connection
- **Dependent**: One affects the other

30. Explain Bayes' Theorem?

Bayes' Theorem is a fundamental concept in probability theory that describes how to update the probability of an event based on new evidence. It provides a way to calculate the **conditional probability** of an event, given that another event has occurred.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Example: Email Spam Detection

Let's say:

- 20% of all emails are spam → P(Spam)=0.20
- 80% are not spam → P(Not Spam)=0.80
- The word "offer" appears in:
  - 70% of spam emails → P(Offer | Spam)=0.70
  - 10% of non-spam emails → P(Offer | Not Spam)=0.10

Now, an email contains the word **"offer"**. What is the probability that it's spam?

Step 1: Use Bayes' Theorem

$$P(\text{Spam}|\text{Offer}) = \frac{P(\text{Offer}|\text{Spam}) \cdot P(\text{Spam})}{P(\text{Offer})}$$

We need to calculate $P(\text{Offer})$:

$$P(\text{Offer}) = P(\text{Offer}|\text{Spam}) \cdot P(\text{Spam}) + P(\text{Offer}|\text{Not Spam}) \cdot P(\text{Not Spam})$$

$$P(\text{Offer}) = (0.70 \cdot 0.20) + (0.10 \cdot 0.80) = 0.14 + 0.08 = 0.22$$

Step 2: Plug in the values

$$P(\text{Spam}|\text{Offer}) = \frac{0.70 \cdot 0.20}{0.22} = \frac{0.14}{0.22} \approx 0.636$$

✅ **Final Answer:**

There's about a **63.6% chance** the email is spam if it contains the word "offer".

31. What is conditional probability?

Conditional probability is the probability of an event occurring given that another event has already occurred. In other words, it is the probability of an event A happening, assuming that event B has already occurred. It is denoted as P(A∣B), which reads as "the probability of A given B."

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Scenario:

Consider a deck of 52 playing cards. Suppose we want to find the probability of drawing a **queen** (event A) from the deck, **given** that the card drawn is a **face card** (event B).

- **Event A**: Drawing a queen.
- **Event B**: Drawing a face card (Jack, Queen, or King).
-
    1. **Probability of event B (drawing a face card):**
        - There are 12 face cards in a deck (Jack, Queen, King of each suit).
        - So, $P(B) = \frac{12}{52} = \frac{3}{13}$.
    2. **Probability of event $A \cap B$ (drawing a queen, which is also a face card):**
        - There are 4 queens in the deck (one from each suit).
        - So, $P(A \cap B) = \frac{4}{52} = \frac{1}{13}$.
    3. **Conditional probability:**

       Using the formula for conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{13}}{\frac{3}{13}} = \frac{1}{3}$$

32. What is the difference between parametric and non-parametric tests?

The difference between **parametric** and **non-parametric** tests lies in the assumptions they make about the data:

Parametric Tests

- Assume the data follows a specific distribution (usually **normal**).
- Require parameters like **mean** and **standard deviation**.
- More powerful when assumptions are met.
  👉 **Examples:** t-test, ANOVA, z-test

Non-Parametric Tests

- **Do not** assume any specific distribution.
- Used when data is skewed, has outliers, or is ordinal.
- More flexible, but less powerful.
  👉 **Examples:** Mann-Whitney U test, Wilcoxon test, Kruskal-Wallis test